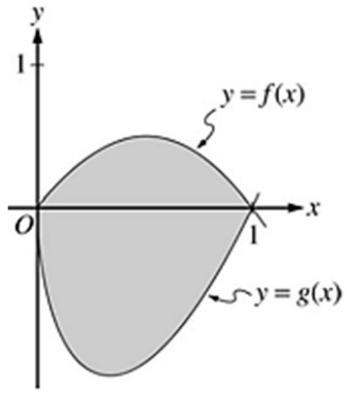
Worksheet 8-4 to 8-6 B Area Between Curves #2

AP Calculus Name_____

Questions 1 – 7 are Calculator Active Questions. A calculator is not allowed on Number 8. Remember, you must set up all answers for calculator active questions to get credit!

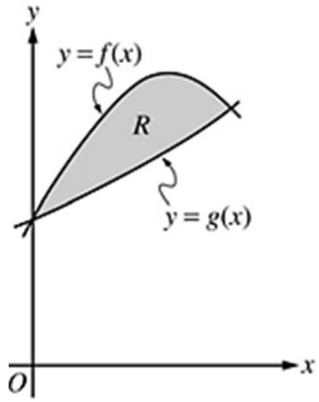
1.) Let f and g be the functions given by $f(x) = 2x(1 - x)$ and $g(x) = 3(x - 1)\sqrt{x}$ for $0 \le x \le 1$. The graphs of f and g are shown in the figure.

(a) Find the area of the shaded region enclosed by the graphs of f and g.



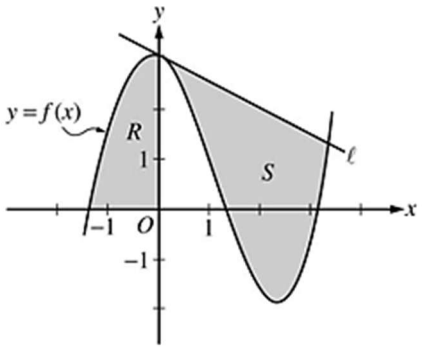
2.) Let f and g be the functions given by $f(x) = 1 + \sin(2x)$ and $g(x) = e^{x/2}$. Let R be the shaded region in the first quadrant enclosed by the graphs of f and g as shown in the figure below.

(a) Find the area of R.



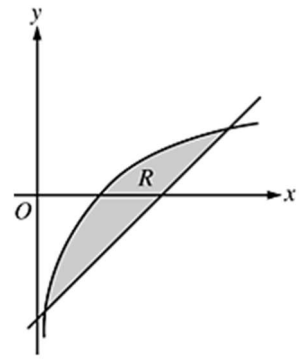
3.) Let f be the function given by $f(x) = \frac{x^3}{4} - \frac{x^2}{3} - \frac{x}{2} + 3\cos x$. Let R be the shaded region in the second quadrant bounded by the graph of f, and let S be the shaded region bounded by the graph of f and line l, the line tangent to the graph of f at $x = 0$, as shown in the graph below.

(a) Find the area of R.

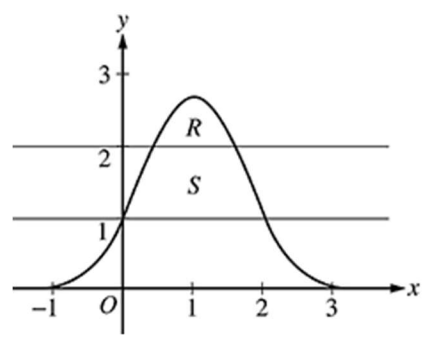


4.) Let R be the region enclosed by the graph of $y = \sqrt{x-1}$, the vertical line $x = 10$, and the x-axis.
(a) Find the area of R.

5.) Let R be the shaded region bounded by the graph of $y = \ln x$ and the line $y = x - 2$, as shown below.
(a) Find the area of R.



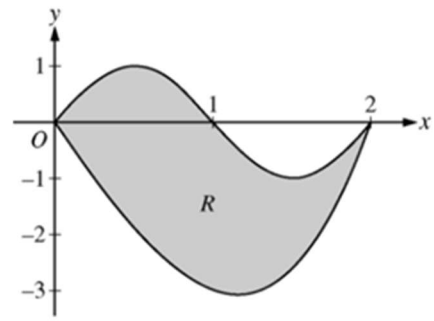
6.) Let R be the region bounded by the graph of $y = e^{2x-x^2}$ and the horizontal line $y = 2$, and let S be the region bounded by the graph of $y = e^{2x-x^2}$ and the horizontal lines $y = 1$ and $y = 2$, as shown above.
(a) Find the area of R.
(b) Find the area of S.



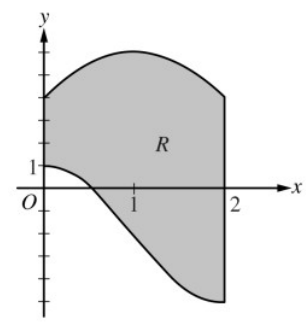
7.) Let R be the region bounded by the graphs of $y = \sin(\pi x)$ and $y = x^3 - 4x$, as shown in the figure below.

(a) Find the area of R.

(b) The horizontal line $y = -2$ splits the region R into two parts. Write, but do not evaluate, an integral expression for the area of the part of R that is below this horizontal line.



8.) **NO CALCULATOR ALLOWED.**

Let R be the region enclosed by the graphs of $g(x) = -2 + 3\cos\left(\frac{\pi}{2}x\right)$ and $h(x) = 6 - 2(x-1)^2$, the y-axis, and the vertical line $x = 2$, as shown in the figure below.

(a) Find the area of R.